Feed Forward Layer

A feed forward layer is a core building block in artificial NNs where information flows in only one direction (from input to output) with no loops.

Characteristics

- **No feedback loops:** Unlike recurrent networks, outputs from the layer are not fed back into itself; data continuously moves linearly toward the final result
- **Key-value memory:** Research shows that transformer feedforward networks act as a massive key-value memory. Keys map to specific textual patterns (the input), and values induce distributions over the output vocabulary.
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In transformers

- **Linear Up-projection:** The input vector x is linearly projected to a higher-dimensional space (e.g., $d_{model} \to 4 \times d_{model}$)
- **Activation Function:** a non-linear activation (commonly ReLU or GELU) is applied, allowing the model to learn complex, non-linear patterns.
- **Linear Down-Projection:** the expanded vector is linearly projected back to the original model dimension to be combined via residual connections.